NAME: _____

BANGWE CLUSTER EXAMINATIONS

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

Thursday, 19 May

CHEMISTRY

Subject Number: M036/II
Time Allowed: 2 hour sessions
8:00 onwards

PAPER II
(40 marks)

Practical

Instructions

1. **This paper contains 6 printed pages. Please check.**
2. This paper contains **two** sections, **A** and **B.**
3. Answer **all** the **four** questions in the spaces provided. The maximum number of marks for each answer is indicated against each question. A pencil should be used for all drawings.
4. Write your **Name** at the top of each page of your question paper in the spaces provided.
5. Do not spend more than **30 minutes** on a single question.
6. In the table provided on this page, **tick** against the question number you have answered

Question Number	Tick if Answered	Do not write in these columns	
1			
2			
3			
4			
	Total		

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SECTION A: (20 marks)

1. Describe how 2000ml of 0.1M HCl can be prepared using a stock solution of 10M HCl.

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_____(10 marks)

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2. With the aid of a well labelled diagram and chemical equations, design an experiment that can be used to electroplate a Silver Spoon with Copper metal. Use relevant equations.

_____(10 marks)

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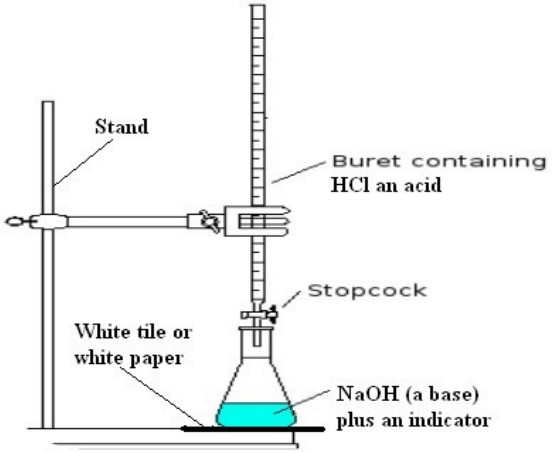
SECTION B (20 Marks)

3. You are provided with a burette or a syringe, a funnel, a measuring cylinder, a beaker or conical flask, a clamp stand, 0.1M Sodium hydroxide (NaOH), Hydrochloric acid (HCl) of unknown concentration, phenolphthalein indicator and a white tile or plane paper.

a. Set up the apparatus as shown below in Figure 1



a. Figure 1

b. Fill the burette with hydrochloric acid (HCl) to the zero mark.

c. Pour 10 ml of Sodium hydroxide (NaOH) into the beaker or conical flask.

d. Add 3 drops of phenolphthalein indicator in the Sodium hydroxide.

e. Gradually add hydrochloric acid from the burette to Sodium hydroxide, shaking the beaker all the time until pink colour disappears.

f. Run the experiment twice or two runs by repeating steps (c) to (e)

g. Fill in the table 1 below. (3 marks)

Number of runs	Initial volume of HCl (ml)	Final volume of HCl(ml)	Volume of HCl used(ml)
First			
Second			

h. Calculate the volume of Hydrochloric acid used (Average titre) (2 marks)

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i. Write a balanced chemical equation for the reaction. (2 marks)

_____

_____

j. Work out the concentration hydrochloric acid solution used. (2 marks)

k. Mention one source of error in this experiment?

_____(1 mark)

4. You are provided with sodium hydroxide (NaOH) pellets, ammonium chloride (NH_4Cl), distilled water, a thermometer, a beaker, a stirring rod, a measuring cylinder, and a spatula.

   a. Pour 50 ml of distilled water into a beaker.
   b. Measure the initial temperature of the water and record it in the table of results.
   c. Add 5 pellets of NaOH to the distilled water and stir until the pellets dissolve.
   d. Measure the temperature of the solution and record it in the table of results.
   e. Repeat steps **a** to **d** using $\frac{1}{2}$ a spatula of NH_4Cl.

Table of Results

SOLUTION	**Initial Temperature (°C)**	**Final Temperature (°C)**
$NaOH_{(s)}+H_2O_{(l)}$		
$NH_4Cl_{(s)}+ H_2O_{(l)}$		

(2 marks)

   f. Draw energy level diagrams for dissolving sodium hydroxide and ammonium chloride in water.
      i. Sodium hydroxide in water:

(4 marks)

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ii. Ammonium chloride in water:

(4 marks)

END OF QUESTIONS